

PRINCIPLES OF FLIGHT

By : 1st Lt. Omar Adel El-Sayed

SECTION TWO

TABLE OF CONTENTS

01

High Lift Devices

- Purpose of HLD
- Concept Behind HLD
- Types of HLD
- Mathematical Representation
- Application in Coding

Purpose of HLD

HIGH LIFT DEVICES

Why Do we need High Lift Devices ..?

Purpose of HLD

- Aircraft are fitted with high lift devices to **reduce the take-off and landing distances.**
- This permits operation at **greater weights** from **given runway lengths** and enables **greater payloads to be carried.**

Concept Behind HLD

How is that Applicable ??

Concept Behind HLD

In Take-off :

For the aircraft to take-off the lift force must equal the aircraft weight

$$L = W$$

$$\frac{1}{2} \rho V_{TO}^2 S C_{L_{\max}} = m \cdot g$$

Concept Behind HLD

$$\frac{1}{2} \rho V_{TO}^2 S C_{L_{\max}} = m \cdot g$$

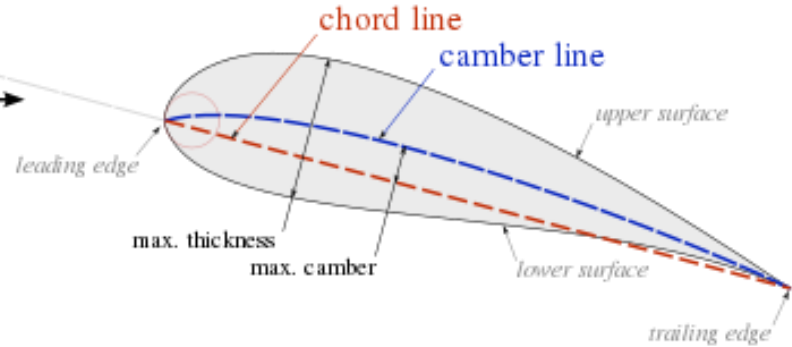


Depends on Airfoil Shape (Camber)

angle of attack

α

relative wind

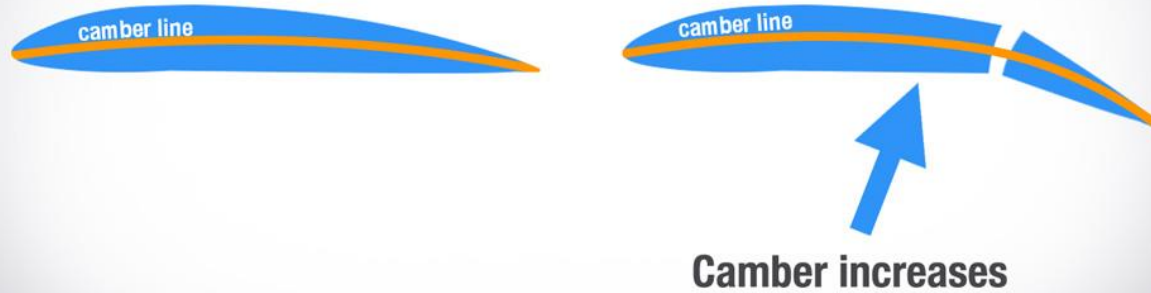


This means that if we can **increase the camber**, we can **increase** $C_{L_{\max}}$

Concept Behind HLD

HIGH LIFT DEVICES

Camber Line Changes With Flaps Configuration



boldmethod ▶

Concept Behind HLD

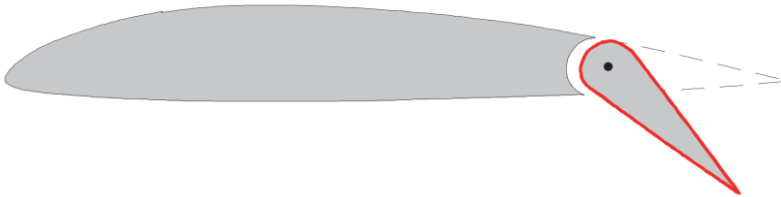
Why not use an airfoil with large camber permanently ?

- It has been shown that increasing the camber of an airfoil section increases the CL at a given angle of attack and increases CL_{MAX} .
- For take-off and landing a cambered section is desirable, but this would give high drag at cruising speeds and require a very nose-down attitude.
- It is usual to select a less cambered airfoil section to optimize cruise and modify the section for take-off and landing by the use of flaps

Types of HLD

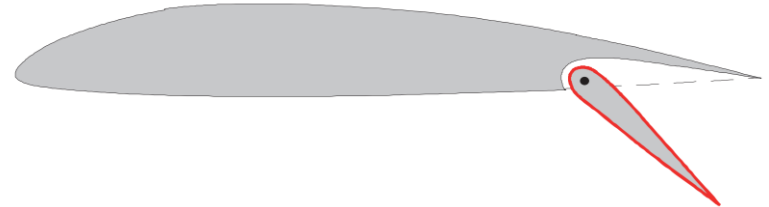
Plain Flap

- The plain flap has a simple construction and gives a good increase in CL_{MAX} , although with fairly high drag.
- It is used mainly on low speed aircraft and where very short take-off and landing is not required.



Split Flap

- The flap forms part of the lower surface of the wing trailing edge, the upper surface contour being unaffected when the flap is lowered.
- The split flap gives about the same increase in lift as the plain flap at low angles of attack but gives slightly more at higher angles as the upper surface camber is not increased, and so separation is delayed.
- The drag, however, is higher than for the plain flap due to the increased depth of the wake.

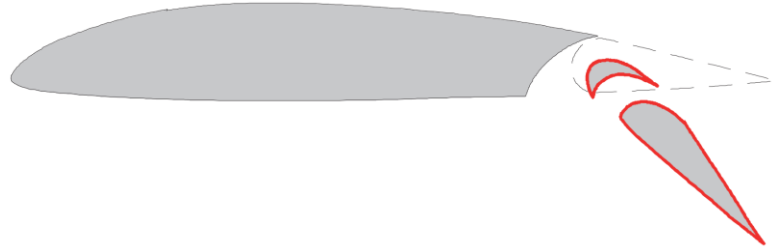


Types of HLD

HIGH LIFT DEVICES

Slotted and Multiple Slotted Flaps

- When the slotted flap is lowered, a slot or gap is opened between the flap and the wing.
- The purpose of the slot is to direct higher pressure air from the lower surface over the flap and re-energize the boundary layer. This delays the separation of the airflow on the upper surface of the flap.
- The slotted flap gives a bigger increase in CL_{MAX} than the plain or split flap and much less drag, but it has a more complex construction.

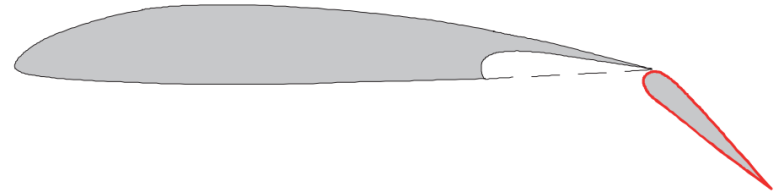


Types of HLD

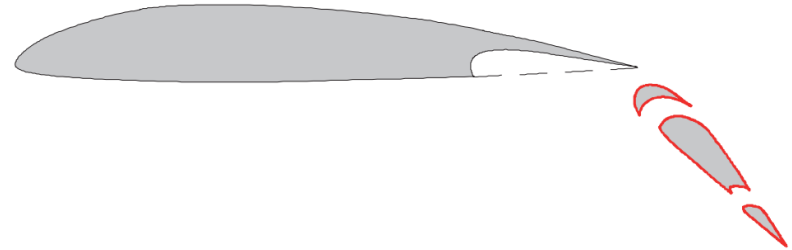
HIGH LIFT DEVICES

Fowler Flap

- The Fowler flap moves rearwards and then down, initially giving an increase in wing area and then an increase in camber. The Fowler flap may be slotted.
- Because of the combined effects of increased area and camber, the Fowler flap gives the greatest increase in lift of the flaps considered and also gives the least drag because of the slot and the reduction of thickness : chord ratio. However, the change of pitching moment is greater because of the rearward extension of the chord.

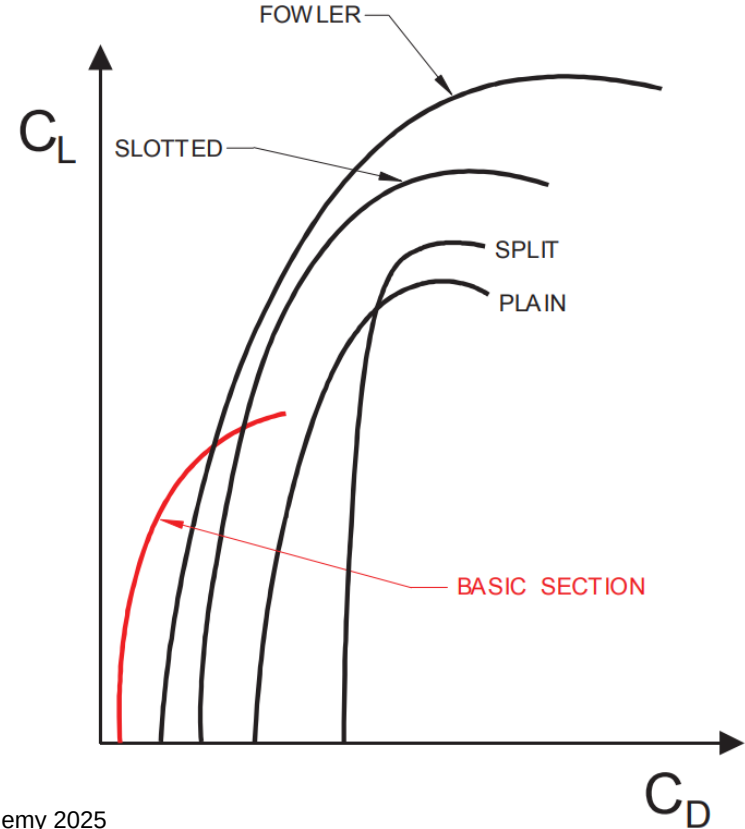
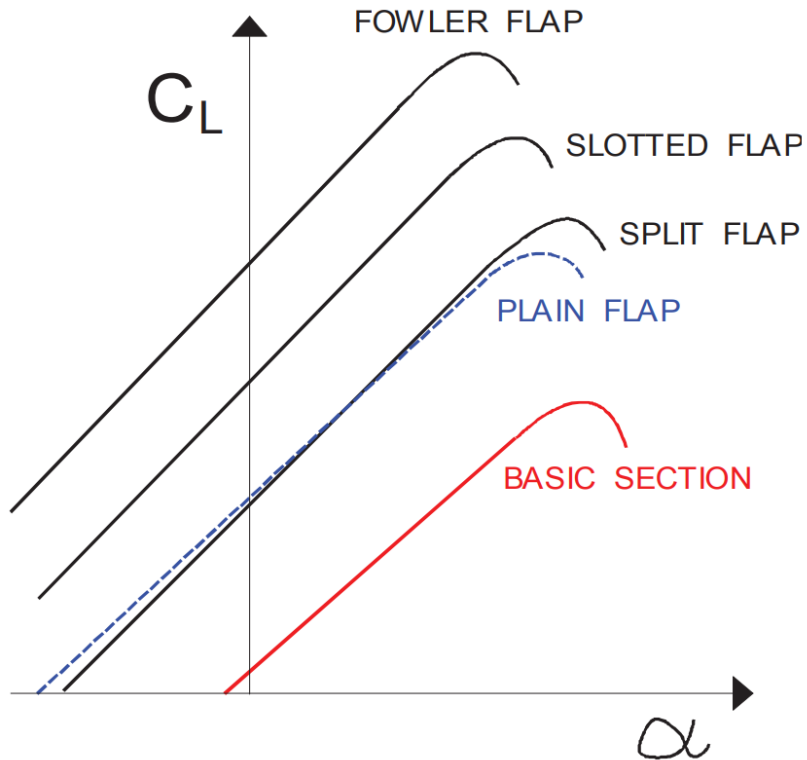


Fowler Flap



Triple Slotted Fowler Flap

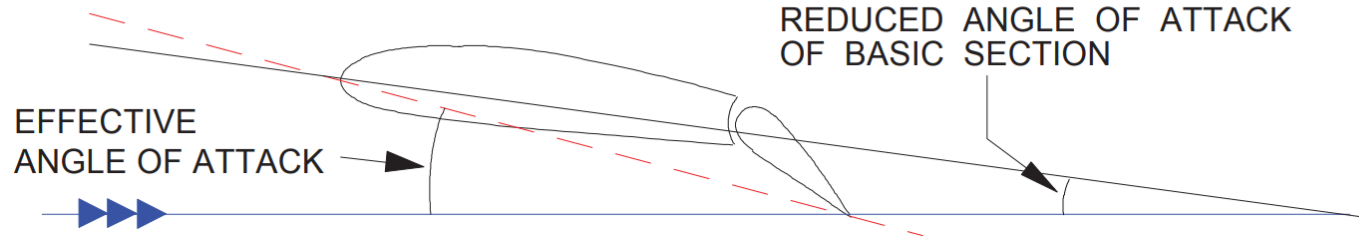
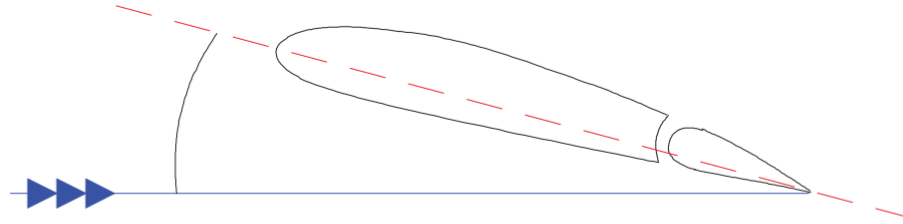
Types of HLD



HIGH LIFT DEVICES

Types of HLD

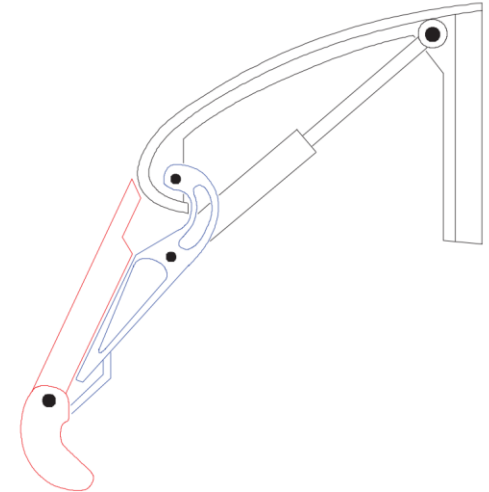
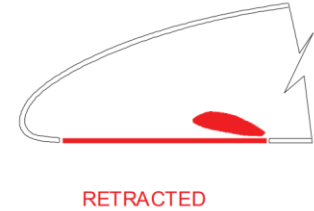
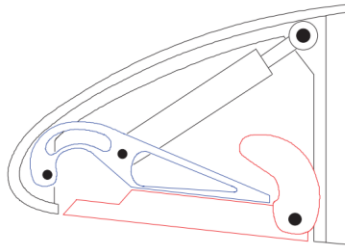
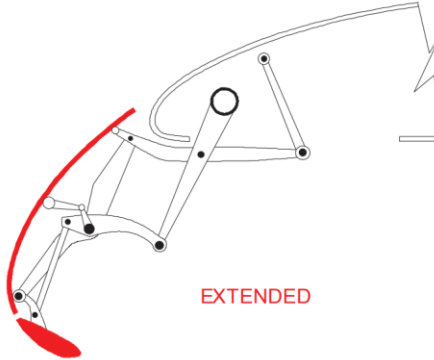
HIGH LIFT DEVICES



Types of HLD

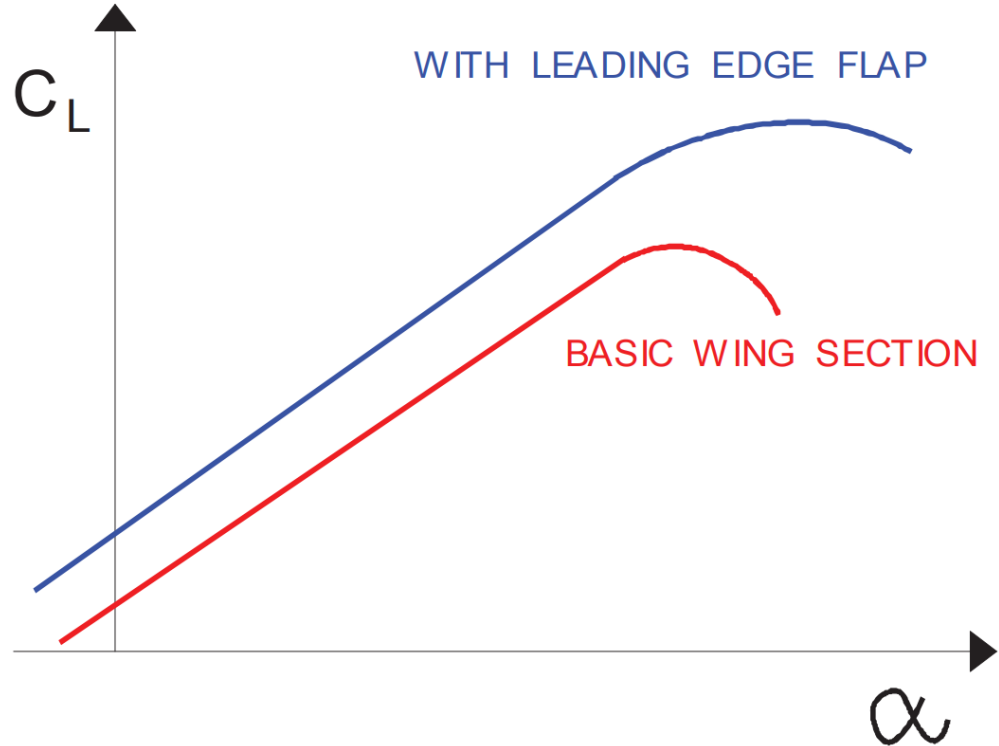
Leading Edge Flap

- On high speed airfoil sections the leading edge may have very little camber and have a small radius. This can give flow separation just aft of the leading edge at quite low angles of attack.
- This can be remedied by utilizing a leading edge flap, which increases the leading edge camber.



Types of HLD

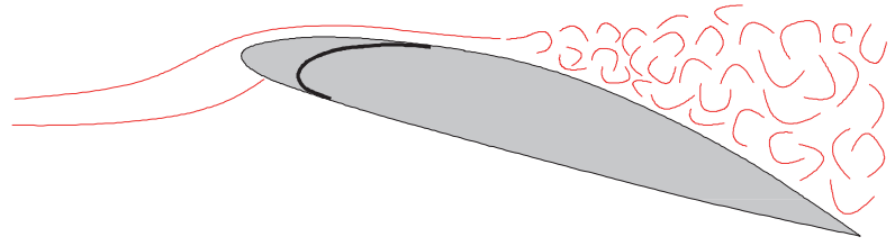
HIGH LIFT DEVICES



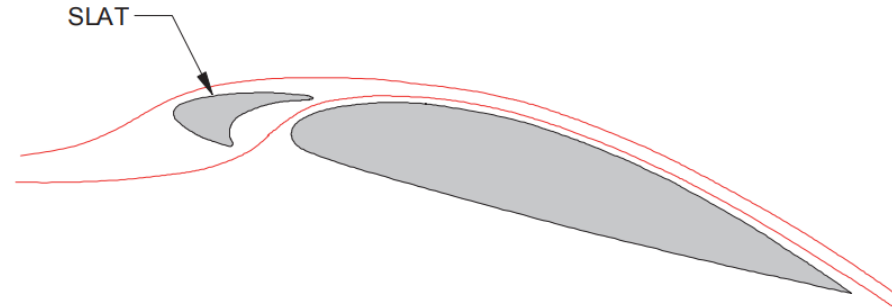
Types of HLD

Leading Edge Slots

- A leading edge slot is a gap from the lower surface to the upper surface of the leading edge, and it may be fixed, or created by moving part of the leading edge (the slat) forwards.
- The slat forms a slot which allows passage of air from the high pressure region below the wing to the low pressure region above it. Additional Kinetic Energy is added to the airflow through the slot by the slat forming a convergent duct.
- **When slats are deployed, the boundary layer is re-energized**

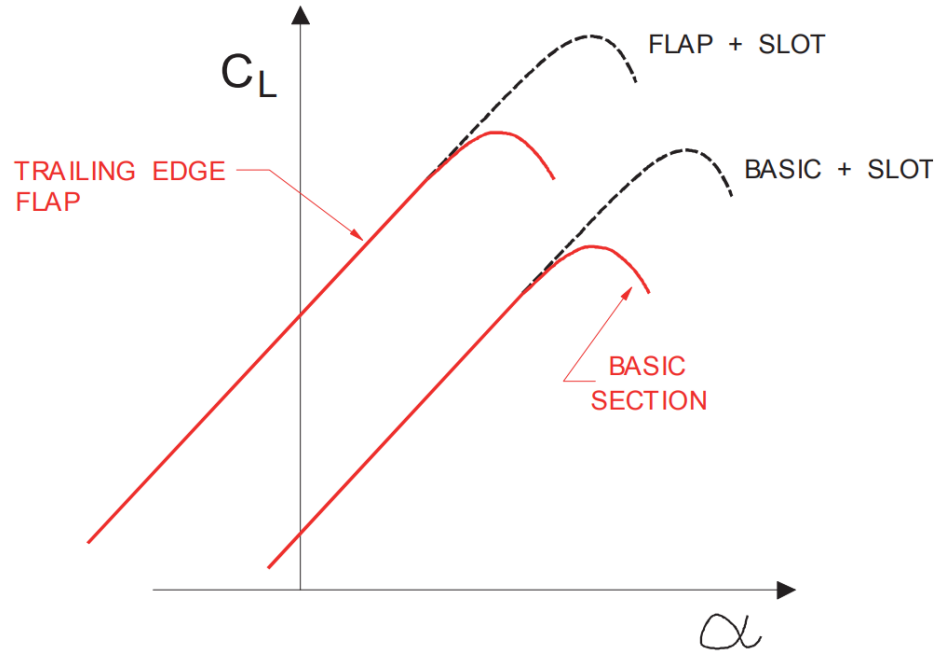
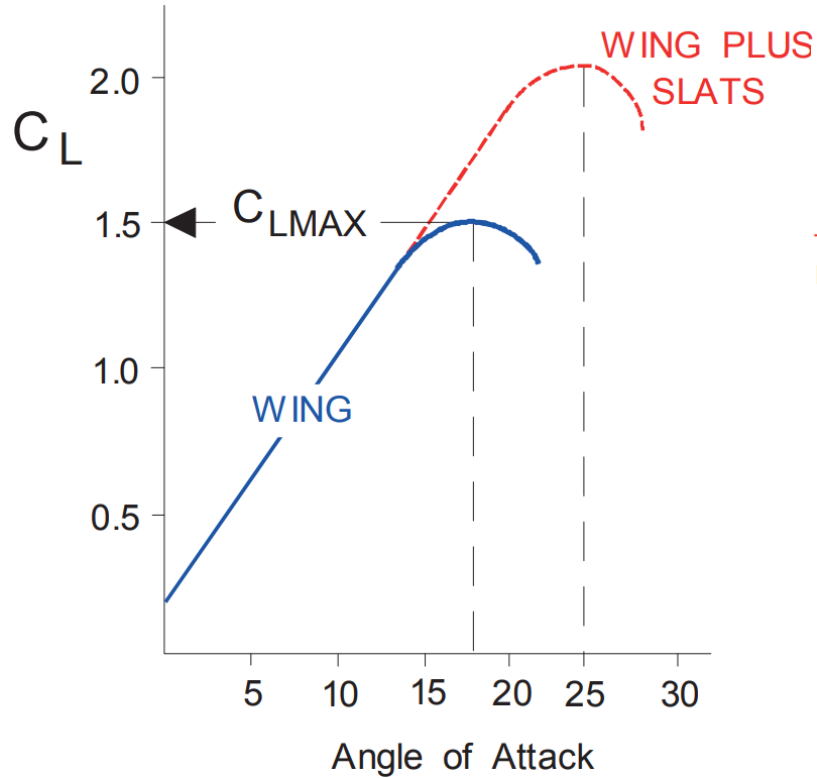


C_{LMAX} WING (Given Adverse Pressure Gradient)



SLAT OPEN - Boundary Layer Re - energized
(Same Adverse Pressure Gradient)

Types of HLD



Mathematical Representation

HIGH LIFT DEVICES

How to Calculate the Lift coefficient for an airfoil ??

Mathematical Representation

For **symmetric Airfoils**, the **Lift Coefficient** can be calculated through this Formula :

Without Flap :
$$C_l = 2\pi\alpha \cdot \left(1 + k \cdot \frac{t}{c}\right)$$

With Flap :
$$C_l = 2\pi(\alpha + \delta) \cdot \left(1 + k \cdot \frac{t}{c}\right)$$

t = maximum thickness of the airfoil,

c = chord length,

k = a constant (typically around 0.77 for many airfoils).

Mathematical Representation

Example : for the **NACA 0012** Airfoil calculate the **Lift Coefficient** at angle of attack **5 deg.**

Solution :

Given Data :

Angle of attack (α): 5°

Flap deflection angle (δ): 5°

Thickness-to-chord ratio (t/c): 12% (0.12)

Empirical constant (k): 0.77 (for thickness correction)

Airfoil type: Symmetric (NACA 0012)

Mathematical Representation

Example : for the **NACA 0012** Airfoil calculate the **Lift Coefficient** at angle of attack **5 deg.**

Without Flap

$$\alpha_{\text{radians}} = \alpha_{\text{degrees}} \times \left(\frac{\pi}{180} \right)$$

Substitute $\alpha = 5^\circ$:

$$\alpha_{\text{radians}} = 5 \times \left(\frac{\pi}{180} \right) \approx 0.0873 \text{ radians}$$

$$C_l = 2\pi \cdot 0.0873 \cdot (1 + 0.77 \cdot 0.12)$$

$$C_l \approx 0.548 \cdot 1.0924 \approx 0.598$$

With Flap

$$C_l = 2\pi(\alpha + \delta) \cdot \left(1 + k \cdot \frac{t}{c} \right)$$

Substitute the values:

$$C_l = 2\pi \times (0.0873 + 0.0873) \cdot (1 + 0.77 \times 0.12)$$

Calculate the effective angle of attack:

$$\alpha + \delta = 0.0873 + 0.0873 = 0.1746 \text{ radians}$$

Use the same thickness correction term (1.0924):

$$C_l = 2\pi \times 0.1746 \times 1.0924 \approx 1.198$$

Application in Coding

HIGH LIFT DEVICES

Write a code to plot the **lift curve** (lift coefficient C_l vs. angle of attack α) for a symmetric airfoil of the NACA 00XX family, both **with and without a flap**, in the angle of attack range of 0° to 10° .

The program should produce a plot showing:

- Two curves: one for the airfoil **without a flap** and one for the airfoil **with a flap**.
- The x-axis should represent the angle of attack (α) in degrees.
- The y-axis should represent the lift coefficient (C_l).